Verification report: the November 1 manuscript in Lean

Confidential. Prepared 2 November 2026 for the author of the manuscript of 1 November 2026 (`nov 1.tex`).

1. Summary

The manuscript contains 64 numbered mathematical statements (theorems, lemmas, corollaries, propositions and claims, including Appendices A and B; Theorem 5.8 is a restatement of Theorem 2.6 and is counted once). Each of them has one final Lean theorem in `LocalBreadth/K/Final.lean`, and each of these 64 theorems is at level K in the sense of your request:

- (K1) independent reviewers compared each statement clause by clause with the tex (hypotheses, quantifiers, definitions and conclusions) and found that it matches;
- (K2) each one is proved by Lean's kernel. `#print axioms` gives only `propext`, `Classical.choice` and `Quot.sound` for all 64 theorems and the 4 auxiliary results listed below. There is no `sorry` anywhere in the project;
- (K3) wherever the Lean representation differs from the manuscript's construction (trees, limits, the topology on histories, the family of the earlier paper), the equivalence is itself a kernel-checked Lean theorem;
- (K4) no statement carries an extra unproved assumption or a weaker reading. Where the manuscript admits two readings, both are proved or the stronger one is proved; Section 5 lists these cases.

The whole project was also replayed module by module by Lean's independent kernel checker (`leanchecker`), and every module was accepted.

All 64 statements reach K. Section 4 gives the Lean theorem for each one.

2. Where the Lean is and how to check it

- Repository: `github.com/ZRTMRH/kleinberg-wei-local-breadth-lean` (private). Branch `nov1-K`, commit `18390f8` (full hash `18390f89cc93db80ccb7a8475a9d11626b11652f`). Lean `v4.35.0` with Mathlib.
- Final theorems: `LocalBreadth/K/Final.lean`, namespace `LocalBreadth.K`. There is one theorem per numbered statement, with a docstring giving the statement number and the tex line.
- The manuscript's own definitions, written as printed:
  - `LocalBreadth/K/PaperForms.lean`: c_n as a limit over R, b_n, b(K) as a limit, the topological b_n with neighbourhoods open in the subspace $\mathscr{H}$_K, the lower Banach density as the printed liminf, and safety and protected sets;
  - `LocalBreadth/K/PaperTree.lean` and `LocalBreadth/K/PaperTreeValues.lean`: the trees of Definition 2.3, with their values.
- How to check:
  1. `lake exe cache get` (downloads the compiled Mathlib);
  2. `lake build LocalBreadth` (9105 jobs; succeeds);
  3. in a file containing `import LocalBreadth`, run `#print axioms LocalBreadth.K.thm_1_5`, and so on for each name in the table;
  4. `lake env leanchecker LocalBreadth.K.Final` (and likewise for any other module) replays the compiled proofs in the independent kernel checker.

Checks run on the final commit.

- `#print axioms` on all 68 declarations of `Final.lean` (the 64 rows plus `dP_eq`, `def_7_5_bridge`, `def_7_14_bridge` and `thm_8_2_F_ord`) gives `[propext, Classical.choice, Quot.sound]` for each. This was rerun on a clean checkout of `18390f8`.
- A text search of the sources for `sorry`, `admit`, `axiom` and `native_decide` finds only the English word ”admit” in comments.
- Kernel replay (`leanchecker`): all 57 modules of the library were replayed one at a time at commit `4958d5b`, and every one was accepted. The single later commit, `18390f8`, changes only `LocalBreadth/K/Final.lean`. That module was replayed again at `18390f8` and accepted.

3. How level K was established

A first formalization of the manuscript already existed. Most of its statements were faithful, but some quantities were represented differently from the tex: suprema instead of limits, trees encoded by addresses, and some choices fixed in advance. To reach K, the following was added. The additions are new files under `LocalBreadth/K/`, and no earlier statement was weakened.

- **Trees of Definition 2.3.** These are countable rooted labelled trees, as printed (`PaperTree`). Lean proves that b_n defined with them equals the earlier representation, in both directions. The Section 7 trees of Definitions 7.5 and 7.14 have their own two-way bridges: `K.def_7_5_bridge` (Final.lean:393) and `K.def_7_14_bridge` (Final.lean:396).
- **Limits as printed.** $c_n(H)$ is the limit as $R \to \infty$, $b(K) = \lim_n b_n(K)$, and the lower Banach density is the liminf over $|R \cap K| \to \infty$. In each case Lean proves that the limit exists and equals the value used in the proofs (`LocalBreadth/K/G3Limits.lean`). The higher-dimensional densities δ_h are connected to their printed limit forms by proved bridge lemmas in the same file.
- **Openness in $\mathscr{H}$_K.** The topological b_n takes neighbourhoods open in the subspace $\mathscr{H}$_K, and Lean proves $\mathscr{H}$_K open (`LocalBreadth/K/N1Open.lean`). This lemma was adapted, with attribution, from your Codex formalization (fan-wei-math/banach-density-hd, commit `33488ac`).
- **Proposition B.1.** It is proved for the family $\mathscr{D}_0$ constructed in the SODA paper itself, for every legal choice of that construction’s parameters. The construction was ported verbatim, with the namespace renamed and attribution in the file header (`LocalBreadth/K/PriorZeroPort.lean`), from your Codex formalization fan-wei-math/banach-density-1d-formalization, commit `327ea0f`.
- **Lemma A.2 and Proposition A.5.**
  - White’s ties are broken by a fixed ordering of E, as in the tex (l.7012, l.7043).
  - Properties (P1)–(P4) and locality are required along the play of that strategy, together with the one-step hypothetical moves used in the proof. This is the reading you stated in your records (DECISIONS.md, Q17–Q20).
  - Claim A.6 provides the potentials for every ordering of E, uniformly in d, ε and τ.
- **Freshness.** Both readings of the global freshness condition are proved where they differ (Section 5). Lean has a transfer lemma, `exists_freshEverywhere_valid` (K/ConventionF.lean:168): any generator that is fresh on full enumerations can be changed into one that is fresh on every input sequence and has the same outputs on every full enumeration.

Review.

- Each section was reviewed by an independent reviewer, who compared every Lean statement clause by clause with the tex. Reviewers formed their verdicts from the tex and the Lean sources before they saw the builders’ claims.
- Every row whose Lean statement was changed in response to a finding was reviewed again, by a fresh reviewer, in its final form. This covers Cor 1.6, Thms 1.5, 1.7, 1.8, 6.1, 6.2, Claim 6.6, Thms 7.1–7.3, Lemmas 7.10, 7.12, 7.15–7.17, Thm 8.2, Lemma A.2, Prop A.5, Claim A.6 and the tree bridges.
- The last review, of commit `18390f8`, found no remaining issue.

4. Table of all numbered statements

All Lean names are in the namespace `LocalBreadth.K`. Line numbers refer to `LocalBreadth/K/Final.lean` at commit `18390f8`. "Both freshness readings" means the theorem states its existence parts both with generators fresh on full enumerations (`FreshOn`) and with generators fresh on every input sequence (`FreshEverywhere`); see Section 5.

Paper statement (tex line)	**Lean theorem (Final.lean line)**	**K verified?**	**Remaining issue / note**
Thm 1.5: one-dimensional characterisation; b_n (combinatorial) = b_n (topological), b(K) = lim b_n is optimal (l.1133)	`thm_1_5` (54)	Yes	None. Both freshness readings. The limit's existence is proved.
Cor 1.6: finite point rank ⇒ density 1/2 by one generator (l.1233)	`cor_1_6` (77)	Yes	None. Both freshness readings. Printed liminf density.
Thm 1.7: in ℤ^d, generators for 1/4 at K_0 and 1/2 elsewhere; x + y ≤ 1/2 (l.1315)	`thm_1_7` (92)	Yes	None. Both freshness readings, for the same X and K_0.
Thm 1.8: main one-dimensional theorem (l.1866)	`thm_1_8` (125)	Yes	None. Both freshness readings.
Claim 2.2: c_n(H) > q ⇒ a bounded exceptional region (l.2067)	`claim_2_2` (147)	Yes	None.
Claim 2.4: every tree has a root-to-leaf path above q < b_n(K) (l.2127)	`claim_2_4` (155)	Yes	None. Trees of Definition 2.3.
Thm 2.5: lower-bound generator (l.2136)	`thm_2_5` (163)	Yes	None. Fresh-on-every-sequence form: `thm_2_5_F` (ConventionFApps.lean:66).
Thm 2.6 = Thm 5.8: b_n equals its topological form (l.2199, l.3116)	`thm_2_6` (173)	Yes	None. Neighbourhoods open in $\mathscr{H}$_K.
Lemma 3.2: finite-rank witness (l.2263)	`lem_3_2` (178)	Yes	None.
Cor 3.3: finite rank ⇒ b(K) = 1/2; one generator for finite CB rank (l.2295)	`cor_3_3` (183)	Yes	None. "The generator of the structural theorem" is stated as "there is a generator"; a reviewer judged these equivalent.
Thm 4.1: upper bound (l.2373)	`thm_4_1` (196)	Yes	None.
Lemma 4.2: race lemma (l.2387)	`lem_4_2` (203)	Yes	None.
Thm 5.1: fixed-ε guarantee (b(K) − ε)_+ (l.2719)	`thm_5_1` (208)	Yes	None. Fresh-on-every-sequence form: `thm_5_1_F` (ConventionFApps.lean:73).
Lemma 5.4: basis of the history topology (l.2855)	`lem_5_4` (216)	Yes	None.
Lemma 5.5: finest history topology (l.2938)	`lem_5_5` (219)	Yes	None.
Lemma 5.6: obstruction closure (l.3023)	`lem_5_6` (222)	Yes	None.
Cor 5.7: safety equals interior (l.3088)	`cor_5_7` (231)	Yes	None.
Thm 5.9: topological main theorem (l.3191)	`thm_5_9` (255)	Yes	None. The limit's existence is proved. Fresh-on-every-sequence form of part 1: `thm_5_9_part1_F` (ConventionFApps.lean:82).
Cor 5.10: protected sets (l.3227)	`cor_5_10` (268)	Yes	None.
Lemma 5.12: boards (l.3382)	`lem_5_12` (285)	Yes	None. Proved for every parent map, which is more general.
Lemma 5.13: forest invariant (l.3498)	`lem_5_13` (289)	Yes	None. Safety as printed. An execution is proved to exist.
Lemma 5.14: uniform board bound (l.3521)	`lem_5_14` (304)	Yes	None.
Lemma 5.15: memory (l.3599)	`lem_5_15` (314)	Yes	None. The records are proved to exist, not assumed.
Lemma 5.16: capture (l.3622)	`lem_5_16` (317)	Yes	None.
Lemma 5.17: validity (l.3663)	`lem_5_17` (320)	Yes	None.
Lemma 5.18: pairing game (l.3685)	`lem_5_18` (323)	Yes	None. The game is encoded as legal runs (`PairingRun.IsLegal`); a reviewer judged the encoding faithful.
Cor 5.19: density on regular boards (l.3719)	`cor_5_19` (326)	Yes	None.
Lemma 5.20: chain of copies (l.3887)	`lem_5_20` (337)	Yes	None.
Lemma 5.21: capture of a fixed copy (l.3939)	`lem_5_21` (340)	Yes	None.
Lemma 5.22: density at a fixed copy (l.4044)	`lem_5_22` (343)	Yes	None.
Thm 5.23: one-dimensional conclusion (l.4085)	`thm_5_23` (353)	Yes	None. Fresh-on-every-sequence form via `thm_2_5_F`, which has the same conclusion.
Thm 6.1: general gap 1/16 (l.4144)	`thm_6_1` (363)	Yes	None. Both freshness readings; the optimum β_h is proved to be the same under both.
Thm 6.2: not simultaneously optimal (l.4167)	`thm_6_2` (367)	Yes	None. One family X and one L_0 for items (1)–(4), under both freshness readings.

Paper statement (tex line)	Lean theorem (Final.lean line)	K verified?	Remaining issue / note
Lemma 6.3: validity commitment (l.4530)	lem_6_3 (370)	Yes	None. Stated over ℤ^d. The ℕ^d form is lem_6_3_Nd (K/Lem63.lean:33).
Lemma 6.4: block (l.4549)	lem_6_4 (373)	Yes	None. ”Distinct” inputs kept as printed. The proof does not use it, so the lemma also holds without it.
Lemma 6.5: trade-off (l.4603)	lem_6_5 (376)	Yes	None.
Claim 6.6: a horizontal test with few outputs (l.4614)	claim_6_6 (379)	Yes	None. The test is a row of a forced Lemma 6.4 block, as defined at l.4611.
Thm 7.1: fixed quota ⇔ protected trees (l.4744)	thm_7_1 (385)	Yes	None. Trees of Definition 7.5 as printed. Both freshness readings.
Thm 7.2: profile achievable ⇔ copy trees (l.4783)	thm_7_2 (389)	Yes	None. Trees of Definition 7.14 as printed. Both freshness readings.
Thm 7.3: finite rank ⇒ 1/2 − ε; no 1/2 on ℤ^d for fast thresholds (l.4803)	thm_7_3 (414)	Yes	None. Part 1 under both freshness readings, with no common rank bound. Part 2 for the target ℤ^d, as printed.
Thm 7.7: topological protected tree (l.5185)	thm_7_7 (434)	Yes	None. Neighbourhoods open in $\mathscr{H}$_K and prefix-closed.
Cor 7.8: obstruction (l.5239)	cor_7_8 (437)	Yes	None.
Cor 7.9: topological protected set (l.5298)	cor_7_9 (440)	Yes	None.
Lemma 7.10: capture (l.5379)	lem_7_10 (443)	Yes	None. Every choice of child occurrence the paper allows. Address encoding of trees (Section 5).
Lemma 7.11: sewing (l.5425)	lem_7_11 (446)	Yes	None.
Lemma 7.12: help (l.5695)	lem_7_12 (449)	Yes	None. Every choice of child, stored strategy and fallback. Address encoding (Section 5).
Lemma 7.13: commitments (l.5840)	lem_7_13 (453)	Yes	None. Item 1 includes the output answering the last committing input.
Lemma 7.15: copy chain (l.6042)	lem_7_15 (456)	Yes	None. Every choice of child occurrence. Address encoding (Section 5).
Lemma 7.16: capture of a copy (l.6059)	lem_7_16 (459)	Yes	None. As for 7.15.
Lemma 7.17: density at a fixed copy (l.6090)	lem_7_17 (462)	Yes	None. Every admissible choice, and every partition with (logsew) and (hsew). Address encoding (Section 5).
Lemma 8.1: local rectangle strategy (l.6202)	lem_8_1 (468)	Yes	None. Proved through the fixed-ordering forms of Prop A.5 and Claim A.6.
Thm 8.2: finite rank ⇒ δ_h with h = h_τ ≥ 1/2 − ε (l.6224)	thm_8_2 (473)	Yes	None. Both freshness readings.
Thm 9.1: no 1/2 for fast thresholds (l.6365)	thm_9_1 (484)	Yes	None.
Lemma 9.2: strip lemma (l.6374)	lem_9_2 (487)	Yes	None.
Lemma A.2: one causal strategy meets every local quota (l.7031)	lem_A_2 (496)	Yes	None. Fixed ordering of E. Properties are required along the prescribed strategy’s play plus one-step hypothetical moves (your Q17 reading); see Section 5.
Claim A.3: discarding small terms (l.7474)	claim_A_3 (499)	Yes	None.
Claim A.4: the quotas suffice (l.7507)	claim_A_4 (502)	Yes	None.
Prop A.5: conditional local rectangle lemma (l.7539)	prop_A_5 (507)	Yes	None. The paper’s quantifier order, with a fixed ordering of E.
Claim A.6: the potentials exist (l.7865)	claim_A_6 (513)	Yes	None. For every ordering of E, uniformly in d, ε, τ, with σ = ε/128.
Prop B.1: $b_n(K_0) = 0$ and $b(K_0) = 0$ for the earlier paper’s family (l.8334)	prop_B_1 (536)	Yes	None in the Lean. Proved for the SODA family $\mathscr{D}_0$ itself, for every legal choice of its stages. Convention for $[\ell]$ and ℕ in Section 5. The citation should be corrected (Section 5).
Lemma B.2: full range of b (l.8421)	lem_B_2 (551)	Yes	None. Fresh-on-every-sequence form: lem_B_2_F (ConventionFApps.lean:152).
Lemma B.3: profile realisation (l.8525)	lem_B_3 (565)	Yes	None. Fresh-on-every-sequence form: lem_B_3_F (ConventionFApps.lean:166).
Claim B.4: α_q ≤ b(L_q) (l.8947)	claim_B_4 (569)	Yes	None. b as printed. Holds for every admissible choice in the construction.
Claim B.5: b(L_q) ≤ α_q (l.9021)	claim_B_5 (574)	Yes	None. As for B.4.

Count: 64 statements, all at level K.

Auxiliary results in the same file (also at K): dP_eq (41), the printed density equals the one used in proofs; def_7_5_bridge (393) and def_7_14_bridge (396), the tree equivalences of Section 7; and thm_8_2_F_ord (400), Theorem 8.2 for generators fresh on every input sequence.

5. Conventions and notes

- **$\mathbb{N}$ and $[\ell]$.** $\mathbb{N} = \{0, 1, 2, \ldots\}$, as the manuscript fixes in Section 6, and $\mathbb{N}^d$ is the nonnegative orthant of $\mathbb{Z}^d$. In Proposition B.1, $[\ell] = \{0, \ldots, \ell-1\}$; the SODA paper does not define this notation. This is the reading under which the root-cover step at l.8355 works when $\mathbb{N}$ contains 0. The alternative $[\ell] = \{1, \ldots, \ell\}$ would make that step fail whenever 0 is not a left endpoint.
- **The function u in Section 7.** l.4724 says $u : [1,\infty) \to (0,\infty)$, and l.5419 says $u : [1,\infty) \to [1,\infty)$. The Lean takes u positive and nondecreasing, as at l.4724. This is the weaker hypothesis, so the Lean theorems are at least as strong as either reading.
- **Freshness.** The manuscript requires $o_t \notin S_t \cup O_{t-1}$ (l.1852). The sentence sits inside the full-enumeration model, so it can be read as freshness on full enumerations of the target. Your records state freshness on every input sequence. Both readings are proved:
  - existence statements are stated in both forms (marked in the table);
  - every "for all generators" statement is stated over generators fresh on full enumerations. That class contains the generators fresh on every sequence, so these statements cover both readings.
- **Lemma A.2 under your reading.** Under the play-states reading, which requires the potential's properties only at the states of the prescribed play plus one-step hypothetical moves, the zero potential satisfies the hypotheses whenever the greedy play already meets the quotas. This is a consequence of the reading, not a defect: in that case the lemma's conclusion is already given.
- **Section 7 trees.** Theorems 7.1 and 7.2 are stated with the trees of Definitions 7.5 and 7.14 as printed. The lemmas 7.10–7.17 are stated for the address encoding of trees, which is proved equivalent to the printed trees: existence in both directions, with labels, protected sets, cutoffs, radii and children preserved. A reviewer noted that there is no single Lean lemma that carries a whole run on a printed tree over to its encoding. That step is a routine relabelling, and it does not affect Theorems 7.1 and 7.2.
- **Citation in Appendix B.** The citation "[KWBanach, Theorem 2.4]" (l.1176, l.8331, l.8337) appears to refer to Theorem 5.1 / Corollary 5.2 of the SODA paper, where the family $\mathscr{D}_\%$ is constructed. This concerns the manuscript text only.
- **Use of your code.** Two components were adapted, with attribution in the file headers, from your Codex formalizations:
  - the openness of $\mathscr{H}_K$, from fan-wei-math/banach-density-hd, commit `33488ac`;
  - the SODA family construction used for Proposition B.1, from fan-wei-math/banach-density-1d-formalization, commit `327ea0f`.